

Markscheme

Mathematics: analysis and approaches
Practice question bank

139. (a)

Consider a complex number z satisfying $|z| = 2$ and $\arg(z) = \frac{5\pi}{6}$.

Find z in the form $a + bi$, giving exact values.

$$z = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \quad (M1)$$

$$= 2 \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) \quad (M1)$$

$$= -\sqrt{3} + i \quad \mathbf{A1}$$

[4 marks]